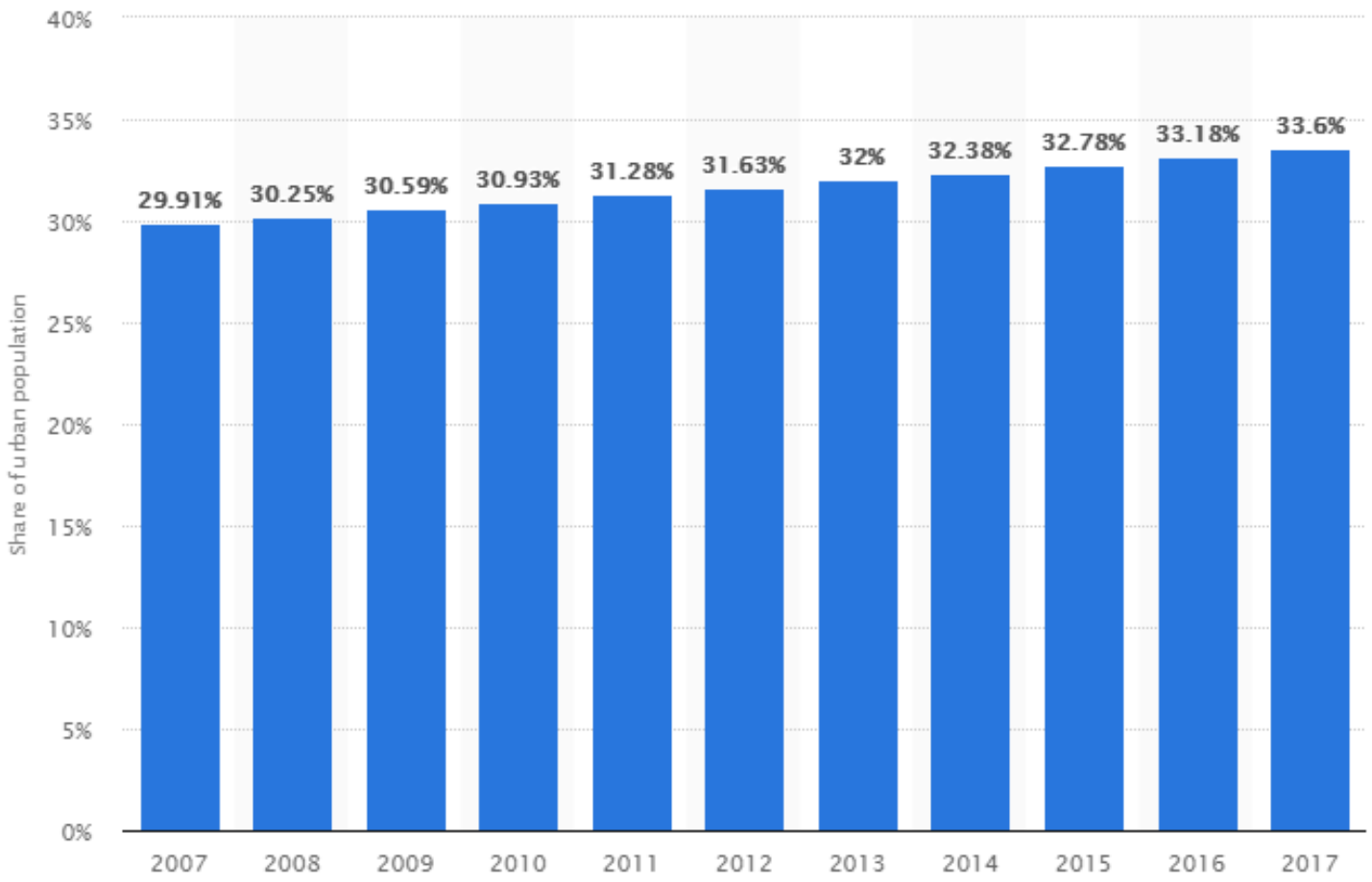
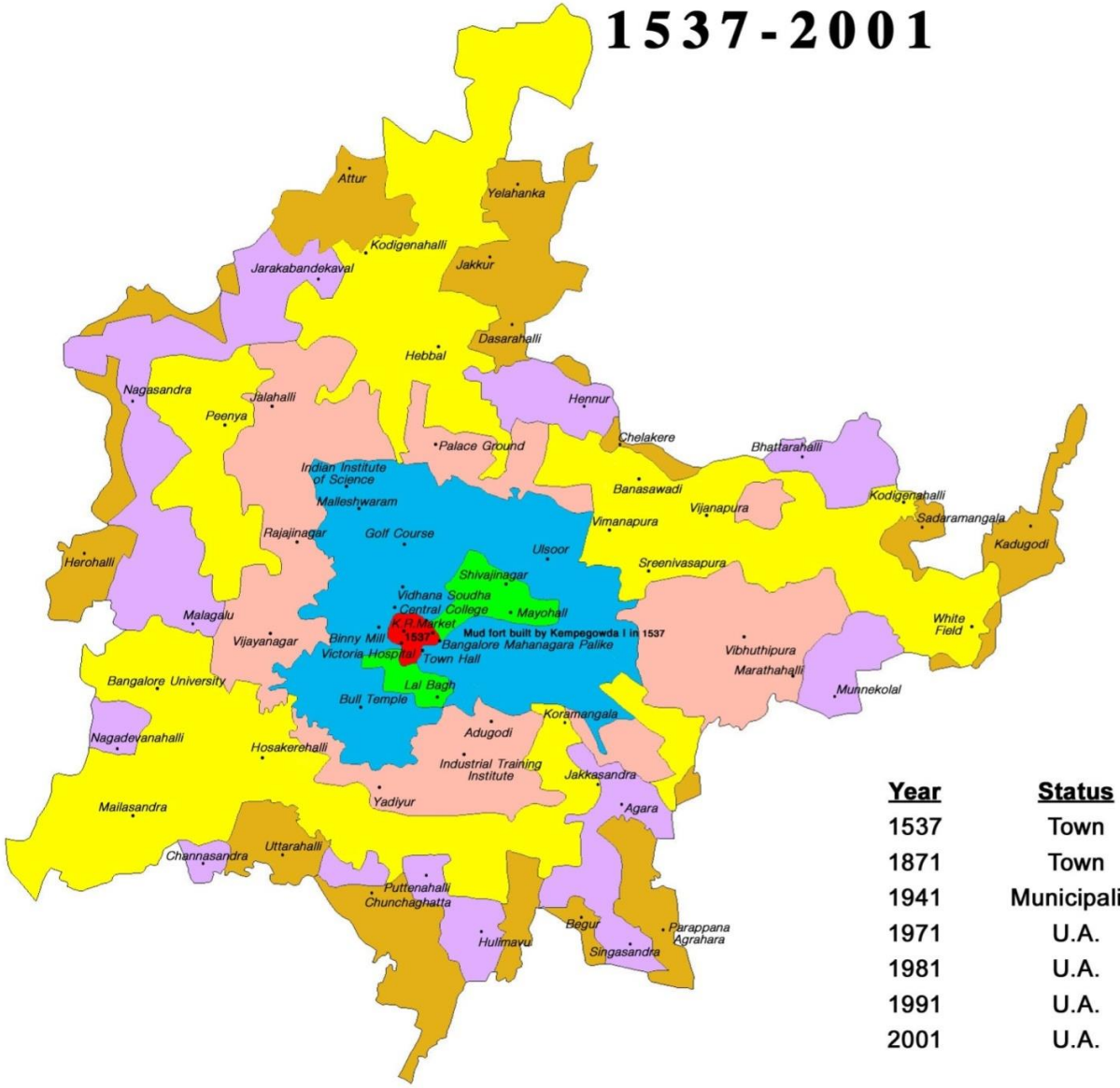
TRANSPORTATION SYSTEMS

India: Degree of urbanization from 2007 to 2017



SPATIAL GROWTH OF BANGALORE
1537-2001

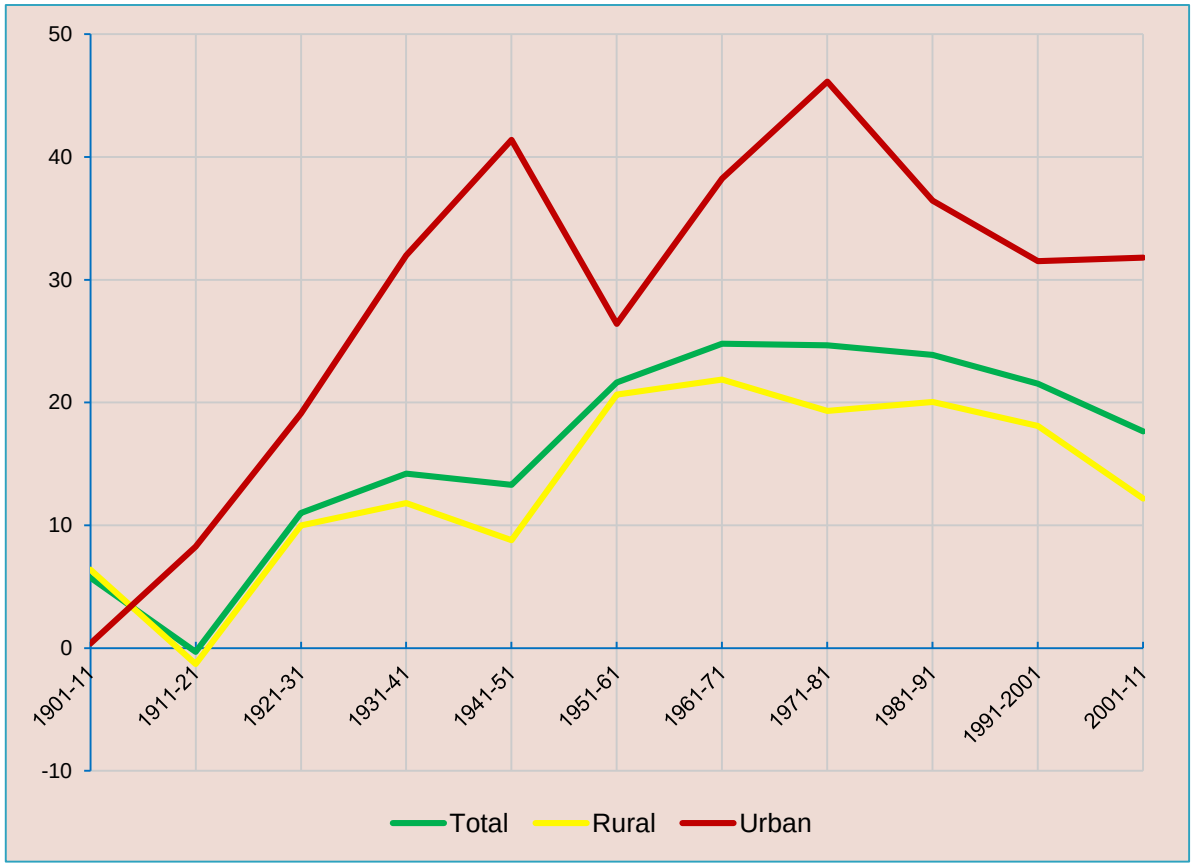


Year	Status	Population	Area in Sq. Km.
1537	Town	**N.A.**	1.85 (approx.)
1871	Town	**144,479**	7.65 (approx.)
1941	Municipality	**410,967**	69.93
1971	U.A.	**1,664,208**	177.30
1981	U.A.	**2,921,751**	365.65
1991	U.A.	**4,130,288**	445.91
2001	U.A.	5,686,844	530.85

U.A. - Urban Agglomeration

Population and Decadal Variation – India 1901-2011

Census Year	Population (in *mn*)		Variation over past decade (in %)	
	Rural	Urban	Rural	Urban
1901	212.54	25.85		
1921	223.23	28.09	-1.29	8.26
1931	245.51	33.46	9.98	19.12
1951	298.64	62.44	8.8	41.4
1961	360.30	78.94	20.64	26.41
1971	439.05	109.11	21.86	38.23
1981	523.87	159.46	19.32	46.14
1991	628.86	217.57	20.04	36.44
2001	742.62	286.12	18.09	31.51
2011	833.09	377.11	12.18	31.80



TRANSPORTATION PLANNING PROCESS

Step 1: Forecasting target year population and economic growth for the subject metropolitan area

Step 2: Allocation of land use and socio-economic projections individual analysis zones according to land availability, local zoning and related public policies

Step 3: Specification of alternative transportation plans partly based on the result of Step 1 and Step 2

Step 4: Calculation of the capital and maintenance costs of each alternative plan

Step 5: Application of calibrated demand forecasting models to predict target year equilibrium flows expected to use each alternative, given the land use and socio-economic projection of Step 2 and the characteristics of the transportation alternatives (step 3)

Step 6: Conversion of equilibrium flows to direct user benefits, such as savings in travel time and travel cost attributable to the proposed plan

Step 7: Comparative evaluation and selection of the best of the alternatives analysed based on estimated costs (step 3) and benefits (step 6)

This information is used to help decision-makers in their selection of transportation policies and programs

Problem Definition

Problem is defined by defining the objective/goal planning. The objective could be-

- To develop a transport system that caters to all sections of society thereby promoting inclusive development
- To develop a transportation system that is primarily sustainable in nature
- To minimize system costs and provide self operation etc.

Solution Generation and Analysis

Based on problem definition, transportation planners have to identify various alternatives and make choice about-

- Various modes like roadway, railway, waterways or airways
- The technological aspect such as high speed train, raised monorails underground transit system, driver information system
- Traffic engineering aspect such as changing or improving flow pattern by making certain road one way, reducing delay on arterial street by improving signalization or grade separated intersection, disallowing certain movement at intersection
- Regulatory aspect such as reserving land for only high occupancy vehicles, disallowing high polluting vehicles, imposing speed limit

EVALUATION OF CHOICES

An alternative has to be evaluated based on following five implications.

- Financially
- Economically
- Land use wise and demographically
- Technologically
- Environmentally

1. **Financial feasibility analysis**

   Financial implications of an alternative are-

   - To determine how much a particular alternative will cost both in term of capital investment and maintenance
   - How the capital cost is structured i.e. whether the investment is phased over a period of time
   - To determine whether it is possible to meet the cost of alternative based on budgetary consideration

2. **Economic impact analysis**

   Here, the objective is to assess various economic effects of an alternative in terms of-

   - Travel time of user.
   - Out of pocket cost of user.
   - Improved business opportunity.
   - Improved accessibility to various locations

3. Technical implication analysis

- Feature of alternative is technically achievable or not
- Technical requirement and their feasibility are subject matter for analysis
- For example provision of high speed train facility between Agra and Delhi is technically feasible because provision of such track and locomotives are possible which can sustain average speed of 125kmph with sufficient no. of wagons

4. Land use and demographical impact analysis

Any transportation facility has number of impact on the land use and demographic characteristics of an area. It may change –

- Land use of an area from unused low land to a bustling shopping complex or large hospital
- Change in land use may often lead to change in demographic statistics of the area
- Better connection between city center and suburbs may lead to shift of high economic group people from city to suburb locations

5. Environmental impact analysis

Construction and use of any major transportation facility have following any or all three direct impact on environment-

- Construction of facility may entail change in the natural feature of land which can lead to problem like loss of habitat of wild life
- Use of facilities (vehicles) may increases air and noise pollution level of immediate surrounding
- A better facility may encourage frequent travel and hence greater use of non renewable fossil fuel.

Dimensions of change

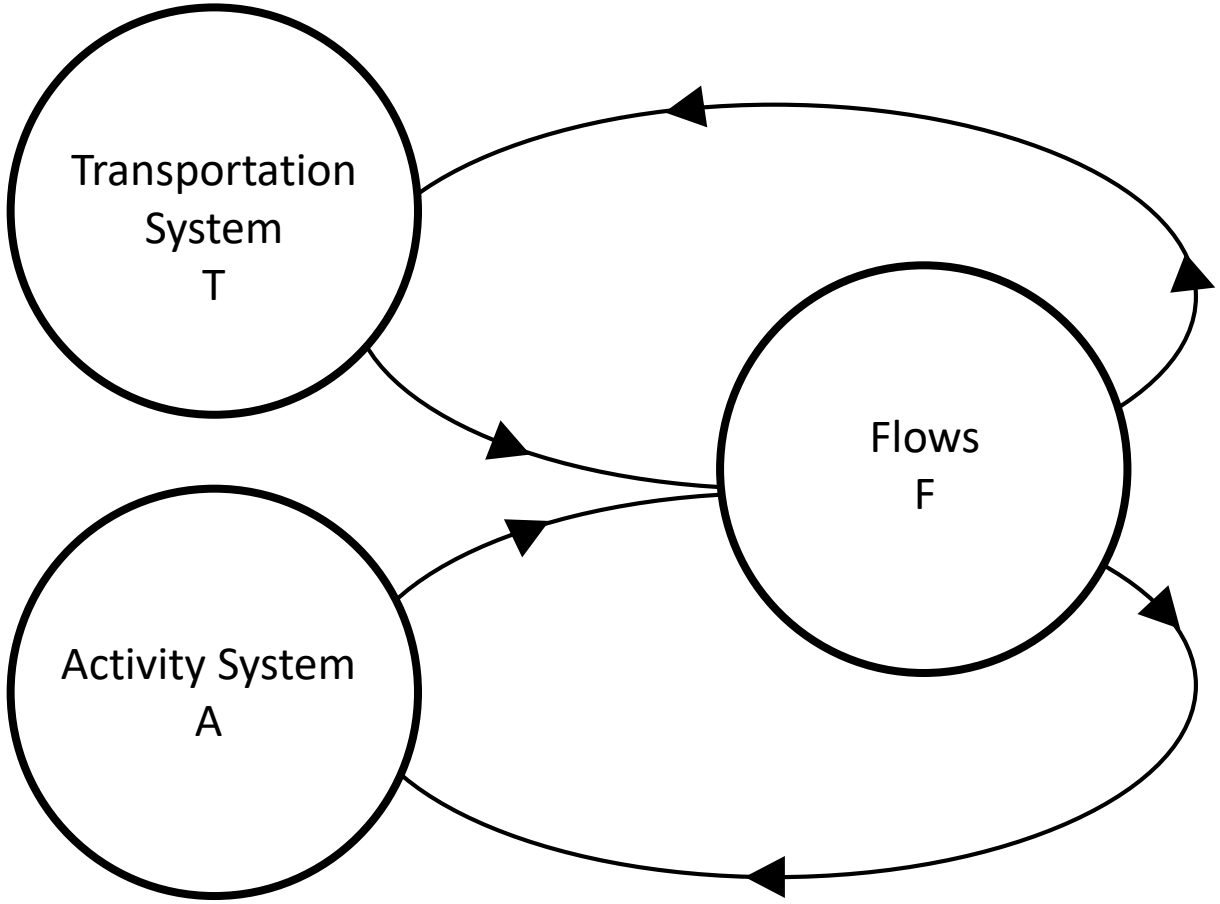
- Demand for transportation: Population, income, land-use
- Technology: metro, monorail, dial-a-ride (Uber/Ola services)
- Values: All users cannot be aggregated in one category– users are divided by socio-economic and demographic categories

Total Transportation System

Analysis of a transportation system should include:

- All modes of transportation
- All elements of the transportation system:
  - Persons
  - Vehicles
  - Network
- All movements of passengers and goods
- Total trip from origin to destination

Transportation and activity systems



Three basic variables

T: Transportation system

A: Activity system: pattern of social and economic activities

F: The pattern of flows in the transportation system– origins, destinations, routes, volume of passengers and goods

The interrelationships between T, A and F

1) The flow pattern (F) in the transportation system is determined both by activity system (A) and transportation system (T)
2) The current flow pattern results in changes over time of the activity system
3) The current flow pattern will cause changes in transportation system

Variables needed to analyse T-A-F system

- Decisions of many individuals, groups, and institutions interact to affect the transportation system (T), activity system (A) and pattern of flows (F)
- Options (or decision variables) are those aspects of the transportation and activity system that can be directly changed by the decision of individuals or institutions. Options can be divided into two categories:
  - Transportation options
  - Activity-system options

Transportation options

- Technology: Metro, bus rapid transit, sensor-based toll collection, compressed natural gas as an alternative to diesel
- Networks: radial network of roads or concentric circles (links and nodes)
- Link characteristics: toll roads, number of lanes, cycle lanes, pedestrian paths
- Vehicles: Auto rickshaws, cycle rickshaws, e-rickshaws

New forms of transport in India

Bus rapid transit in Delhi

Delhi Metro

Mono rail
Mumbai

E-rickshaw

Activity-system options

- The users decide what modes to take for their origin and destination
- The operator of public transport system decides the routes and the frequency of the service
- Industries decide whether to use road-based or rail-based freight transport
- Town Planners decide where to develop markets, industries and residential areas
- Policy makers decide taxes, subsidies and other financial matters

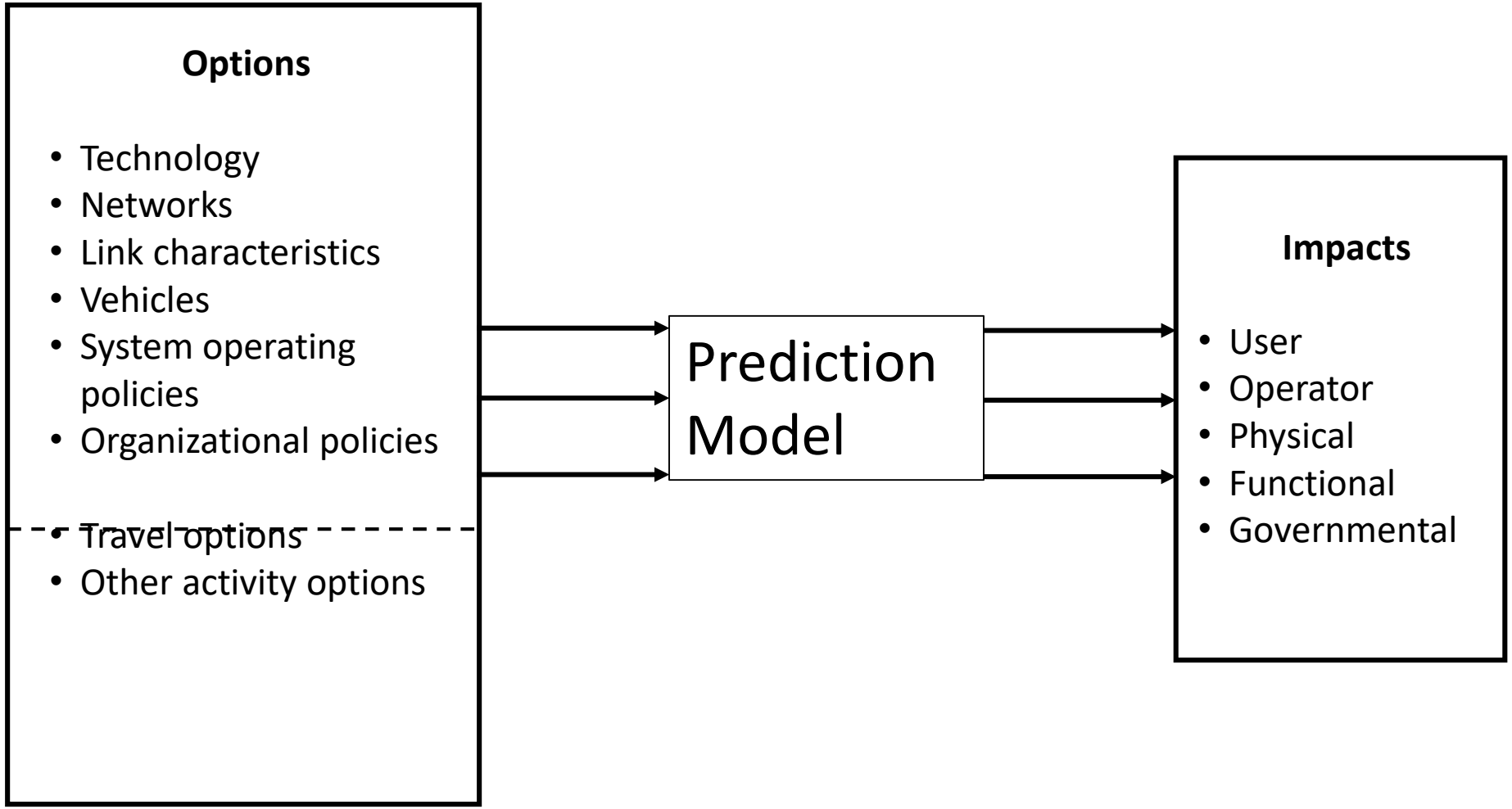
The consequences of transportation

- The evaluation of alternative transportation system requires consideration of all relevant consequences
- The impact categories can be divided into following:
  1. User impacts: Impacts on travelers and shippers of goods (industries), such as travel time delay, lack of transportation options, cost of travel, inconvenience
  2. Operator impacts: Public transport operators, airport authorities, railway authorities, highway maintenance authority

The consequences of transportation

- Impact Categories (cont'd)
  3. Physical impacts: Such impacts caused by physical presence of transportation system affect even the non-users, such as displacement of population to construct highways, noise, vibration (next to railways), air pollution
  4. Functional impacts: The impacts on activity system as users change their travel patterns in response to transport system
  5. Governmental impacts: Depending on the location and the type of government agencies, the affect groups include municipality, state government or central government

Prediction of flows

- The core of transportation system analysis is the prediction of changes in flow
- The challenge is to determine the change in flow, when transportation system and activity system changes

The prediction problem